

System Design Exercise



Food Deliveries R' US (FDRS) wants to launch a food delivery app to compete with market leaders like UberEats, DoorDash, and GrubHub. We need your assistance to design a system that can compete with them and serve its purpose by delivering food in a contactless manner.

Functional Requirements:

Customers are able to search for restaurants, browse menu, place order, track status, and pay for orders. Restaurants are able to create menu, receive notifications of the order placed, drivers assigned, status of an order etc. Drivers are able to receive notifications for orders in the area, assign orders to them, and check for order status when it is available for pickup.

Quality Attributes:

We need a scalable system that is able to cater to 1M unique daily customers across the nation. The plan is to onboard 100K restaurants each having 15+ menu items per restaurant. The latency and uptime of the system are really critical to us.

Constraints:

Java, Microservices, Cloud, and Open Source technologies are the choices made by the CTO.

Principles:

Any core principle that you really feel passionate about. Please clearly state any assumptions that you'd like to make.

Hint: Take some time to think over the problem, ask any questions that will help to crystalize your thoughts. We are not really looking for a complete production system but rather your approach to solving this problem and the solution that you can present to the panel.